



Determinant of Adaptation Strategies to Climate Change by Cassava Farmers in Anambra State, Nigeria

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ABSTRACT

Determinant of choice of adaptation of climate change by cassava farmers in Anambra State, Nigeria was studied using 120 respondents. Primary data were collected through use of structured questionnaire and oral interview. Percentages responses and multinomial logit model were used to address the objectives of the study. The result of adaptation strategies to climate change used by cassava farmers were diversification, use of resistant varieties and early planting. Also, the result of the multinomial logit model analysis showed that the coefficients of farmers' farming status, farming experience and membership of organization were significant and positive to choice of adaptation strategies. Also, the barriers to adaptation to climate change in the study area were poor access to credit, lack of information, high cost of climate change adaptation and high cost of inputs. The need to enhance farmers' access to credit facility, encourage farmers to join or form cooperatives, diversification of crops in their farms and use of resistant varieties were recommended.

Introduction

Agriculture is an important sector of the Nigeria economy. Agriculture is source of food security, employment, contributes to GDP, raw materials for industry, and among others functions (Ume and Kaine, 2021). Among arrays of crops grown in Nigeria, cassava stands prominent (National Root Crop Research Institute, (NRCRI), 2018). Cassava (*Manihot esculenta* Crantz), a food and cash crop in the tropics and many countries in Africa. The crop contributes significantly to the diets of over 800 million people, with per capita consumption averaging 102 kilograms per year. (Okoye, *et al*, 2021). The authors further stated that in some areas in Africa that the crop constitutes over 50 percent of the daily diets of the people. Nigeria is the highest producer of about 54.8 million metric tonnes in world (Nzeakor and Ume, 2021).

Cassava is grown in all agro- ecological zones of Nigeria but production is highest in the North-Central, South East and South regions by majorly smallholder farmers (Ume and Mbah, 2022). Cassava is rain fed and its production is vulnerable to climate change. Climate change is change in climate which is attributed directly or indirectly to human activities that alter the composition of the global atmosphere and which are in addition to natural variability observed over comparable time periods (Deressa, *et al*, 2008; Ozor and Cynthia, 2010). The incidence of climate change can manifest in form of increase in post-harvest losses due to climate variability, flooding, high rainfall intensity, shortening of crop cycle, crop destruction in the field by rain and wind, drought, decrease in soil moistures and loss of farm land/households to floods and erosion (Ozor and Cynthia, 2010).

The effects of climate change to different sectors of the economy such as agriculture, health, water resources and energy among others are well acknowledged. Climate variability has negative influence on the household members' nutrition and consumption habit as result of fluctuation of prices of the food stuff, which could be orchestrated by low production (Mohammed, 2014). Additionally, climate change could lead to increased food insecurity through crop destruction in the field, post-harvest losses, loss of arable land and increased growth of weeds as result of abnormal rainfall and drought (Deressa, *et al*; 2008).

Farmers have developed adaptation methods to climatic changes over decades. Adaptation according to Ume and Kaine, (2017) refers to modifications in human - environmental systems in response to observed or anticipated variation in climatic stimuli. In the same vein, adaptation is the process of improving individual or society's capability to deal with changes in climatic conditions across time scales (Mohammed, 2014). Some of the adaptation strategies by the farmers, included use of resistant varieties, crop diversification, prompt weeding, contour cropping across hill slopes, expansion of cultivated areas, adoption of zero or minimum tillage, contour bund, cultivation of marginal lands, abandonment of crop land, lengthened fallow, use of weather forecasts and irrigation schemes (Shongwe, *et al.*, 2014). Adaptation to climate change requires that households first appreciate that climate has changed, identify useful adaptation options and choice among a wide range of adaptation strategies. Deressa, *et al* (2008) on the other hand found that access to credit, extension information, amount of seasonal rainfall, geographical location, household size, age and literacy of household head were the determinants of farmers' choice of adaptation methods to climate change.

The study was specifically conducted to: describe the socioeconomic characteristics of the respondents, identify the adaptation strategies to climate change, determine the choice of adaptation to climate change and identify the barrier to adaptation / choice of adaptation strategies in the study area.

Methodology/Materials and Methods

Study Area

Anambra State is located between longitude 6° 36'E to 7° 21' of Greenwich Meridian and latitude 5°38'N to 6° 47'N of Equator. The State is bounded in the North by Kogi State, in the west by River Niger and Delta State, in the South by Imo State and on the east by Enugu State. It has twenty one (21) Local Government Areas with Awka as the State Capital. The state is divided into four agricultural zones of Aguata, Anambra, Awka and Onitsha. The zones are further delineated into 24 extension blocks and 120 circles. Farming is one of the main occupations of the people, majority of who are small holders. The study area is also well known for its commercial activities ranging from trading and crafts among others as could be seen in Awka, Nnewi and Onitsha. The rural communities are involved in agricultural activities. Crops grown include but not limited to: yam, cassava, rice, maize, cocoyam, cowpea, tomatoes and vegetables, while the livestock produced in the state include poultry, sheep, goats and to some extent pig. The people also engage into other economic activities apart from farming, included trading, saloon, automobile, vulcanizing, bricklaying, tailoring and among others.

Sampling Techniques

Multistage random sampling techniques was used to select the respondents that were used for the study. First involved the selection of Agricultural Zones. Three out of the four agricultural zones were randomly selected. They were Anambra, Aguata and Awka. The second stage involved the selection of extension blocks. Two extension blocks were randomly selected from each agricultural zones and two (2) circles were further selected

from each of these blocks. Stage three (3) involved the selection of farmers. Ten farmers were randomly selected from each of the circle giving a total sample size of 120 farmers z study.

A structured questionnaire and oral interview were used to elicit information on primary data.

Method of Data Analysis

The objectives of the study were analyzed using percentage responses and multinomial logistic model.

Model Specification

The multinomial logit (MNL) model was used to access the determinants of cassava farmers' choice of adaptation strategies in the study area.

The MNL Model is stated implicitly as follows:

$$Y_i = \ln(P_i, PI) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \beta_6 X_6 + \beta_7 X_7 + \beta_8 X_8 + \beta_9 X_9 + e_i \quad (3)$$

Where

Y_i = adaptation strategy (no adaption; adoption of zero or minimum tillage, use of resistant crop, use of organic manure, changes in planting dates, changes in harvesting dates, mulching/ use of cover crops, cultivation of marginal lands, , lengthened fallow, changes in the timing of land preparation activities and irrigation).

X_i , where $i = 1, 2, 8$ are explanatory variables,

X_1 = Gender of the farmers (male=1, otherwise=0),

X_2 = Age of the farmers (years),

X_3 = Educational attainment (years),

X_4 = Experience in farming (years),

X_5 = Access to extension Services (access=1, otherwise=0),

X_6 = Membership to social network (member=1 otherwise=0),

X_7 = Access to credit (access=1, non-access =0),

X_8 = Farming status (Full-time; 1 and part-time; 0)

e_i = error term

Results and Discussion

Table 1: Distribution of Respondents According to Socioeconomic Characteristics

Factor	Frequency (480)	Percentage	Mean
Age in Years			
20 – 40	32	26.67	
41 – 61	88	73.33	
Mean			42 years
Years of Farming Experience (yrs)			
1 -10	22	18.33	
11 – 20	30	25.00	
21 and above	68	56.67	
Mean			26 years
Access to extension services (dummy)			
Access	37	30.83	
No Access	3	69.17	
Membership of Organization (Dummy)			
Yes	84	70.00	
No	36	30.00	
Educational Level (Years)			
No formal education	24	20.00	
Primary	66	55.00	

Secondary	20	16.67
Tertiary	10	8.33
Mean		7 years
Access to credit use (Dummy)		
Access	82	68.33
Non-Access	38	31.67

Source; Field Survey, 2022

Table 1 showed that 73.33% of the farmers were within age range of 41 – 61 years, while 26.67% were within the age bracket of 20 – 40 years with a mean age of 42 years. The result indicated that the respondents were in their active age of production. This could afford them the opportunities of making rational decision making on choice of adaptation strategies to climate change to make (Deressa; *et al*, 2008). The analysis of the years of farming experience showed that sixty eight (68) (56.67%) of the farmers had farming experience of above 21 years with a mean farming experience of twenty six (26) years. This inferred that the farmers were well experienced. The more experienced a farmer is, the higher the probability of the farmer having enough practical knowledge on how to overwhelm the intricacies of choosing the appropriate adaptation strategies to guide their cassava farms against havocs of climate change.

The result of access to extension service as indicated by the farmers showed that 30.83% of the respondents had contacts with extension agents, while 69.17% had no contact with extension agents. This implied a poor extension outreach, hence depriving the farmers the extension services such as technology information dissemination and technical assistants that could have affirmed the opportunities of making right choices of adaptation strategies options (Mohammed, 2014 and Denkyirah, *et al.*, 2017). Further analysis of membership of organization indicated that 70.00% of the respondents affirmed to be members of cooperatives, while 30.00% were not. Cooperative through training could guide he members of choice of best adaptation options to overcome the terrors of climate change. Ozor and Cynthia, (2008) made similar assertion. They opined that through crossbreeding of ideas among member, solutions to their members could be met, including choice of climate adaptation strategies.

Analysis of the educational attainment of the farmers as indicated in Table 1, revealed that 20.00% of the respondents had no formal education, while 80.00% had formal education. Educational attainment enhances the adoptability and receptiveness of individuals especially that could improve their welfare such as checkmating the hazards in their cassava farms as result of climate change among others (Hassan and Nhemachena, 2008). Uloh *et al.*; (2023) opined that educated people have more information seeking behavior to overcome challenges such as choice of climate change adaptation strategies. The result implied that the farmers were well educated.

Table 1 also showed that 68.33% of the farmers had access to credit, while only 31.67% had no access to credit. Credit access enhances the farmers the opportunity of information to guide on making right choice of adaptation strategies to climate change in cassava farming. The finding of Ume and Kaine, (2017) concurred to the assertion. They reported that credit access aids farmers in procuring inputs at right time in abating climate change.

Table 2: Adaptation Strategies by Cassava Farmers

Strategies	Frequency	Percentage (%)
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i.	Use of resistant varieties	100	91.67
ii.	Diversification	110	83.33
iii.	Irrigation	46	38.33
iv.	Early planting	86	71.67
v.	Minimum or zero tillage	64	53.33
vi.	Cultivation of marginal lands	68	56.67
vii.	Mixed cropping	72	60.00
viii.	No adaptation	24	20.00

Multiple Responses

Source: Field Survey, 2022

Strategies adopted by cassava famers in the study area was determined and presented in Table 2. The result revealed that 91.67%) of the respondents embraced the use of resistant varieties. The result among other variables showed that 83.33% implemented crop diversification in contending climate change in their cassava farms in the study area. Crop diversification refers to the addition of new crops or cropping systems to agricultural production on a particular farm taking into account the different returns from value added crops with complementary marketing opportunities. The aim of crop diversification was to increase crop portfolio so that farmers were not dependent on a single crop to generate their income. When farmers go for single crop type they were likely to be exposed to high risks in the event of unforeseen climate events that could severely impact agricultural production, such as emergence of pests and the sudden onset of frost or drought (Mabe *et al* 2014). The result also showed that 38.33% of the farmers used irrigation to mitigate the effect of climate change. A further analysis indicated that 71.67% of the farmers adopted early planting strategy while 53.33%, 56.67% and 60.00% implemented minimum or zero tillage, cultivation of marginal lands and mixed cropping respectively. It was also revealed that 20.00% had no adaptation strategy. This implied that the farmers were likely to be prone to crop failure and economic crunch resulting from climate change (Ume, and Kaine, 2017).

The multi nominal Logit model result of factors influencing farmers' choice of adaptation was determined and presented in Table 3. The likelihood ratio statistics indicated by statistics π^2 (182.877) and was highly significant, suggesting that the model had a strong explanatory power. The coefficient of age) had a negative and significant impact on the choice of adaptation strategies to climate change using adoption of zero or minimum tillage -1.2654(0.6551), early planting -2.0057(0.1156), mixed cropping -2.0054 (0.8833) and cultivation of marginal lands .0956 (0.1151) could be high. This result is not inconsonance with the findings of Deressa, *et al*; (2008) and Mabe; *et al*; (2014). They asserted that as the farmer advances in age, the more conservative he or she will be in adopting new innovation, no matter how economic feasible the technology will be for fear of failure.

Coefficient of education 0.1264 (0.0154determined showed that the variable significantly influenced adaptation strategies to climate change.) Farmers with many years of farming experience have interacted much more with the climate in relation to their activities and therefore, have good knowledge of environmental factors as they relate to their daily operations (Ume and Kaine, 2017 and Adejumo, *et al*, 2014).

The coefficient of farming experience 3.0334 (0.2252) was positive and had significant effect on adaptation strategies to climate change. This implied that the more farmers are

experienced on climate change, the more likelihood of using adaption measures such as zero or minimum tillage, diversification, diseases resistant varieties and irrigation in abating climate change.

The result also showed that membership of organization 3.3314 (0.0040) significantly influenced adapting to climate change using of zero or minimum tillage, use of diseases resistant varieties, early planting, diversification, cultivation of marginal lands, mixed cropping and irrigation. Evidence had shown that farmers joining one form of organization or the other such as cooperative through interaction could aid in making best decision options among others. The finding of Al-Hassan, *et al* (2013) concurred to above assertion.

The farming status 2.0652 (0.0013) determined, found that it significantly influenced the choice of adapting to climate change using zero or minimum tillage, disease resistant varieties, diversification and irrigation. This implied that a full time cassava farmer can avert the risk emanating from climate change

Table 3: Multi nominal Logit model result of factors influencing farmers' choice of adaptation

Variables	Minimum or zero tillage	Resistant diseases varieties,	Early planting	Diversification	cultivation of marginal lands	Mixed cropping	Irrigation	No adaptation
Age	-1.2654*(0.6551)	-1.0054*(0.4456)	- 2.0057**(0.1156)	0.5457(0.3354)	-1.0956*(0.1151)	- 2.0054**(0.8833)	0.2204(0.5201)	0.7754(0.1711)
Education level	0.1264(0.0154)	0.2152(0.0059)	0.0256(0.2771)	0.0609(1.3300)	0.0688(0.2466)	0.0694*** (0.6758)	0.0194(0.0607)	0.7254(0.3351)
Experience	3.0334*** (0.2252)	4.0346*** (0.0632)	2.0088** (0.0239)	4.6624*** (0.7108)	0.0269(0.4118)	0.0190(0.0052)	4.0270*** (0.0398)	0.5519(0.2281)
Extension Services	0.0092(0.0366)	0.0058(0.0173)	0.2188* (0.9212)	0.0559** (0.0257)	0.6652** (0.0155)	0.2900(0.9210)	0.0037(0.0996)	0.5513(2.0773)
Organization	3.3314*** (0.0040)	3.0996*** (0.3301)	2.3460*** (0.6602)	3.4409*** (0.7700)	5.11008*** (1.3392)	4.7763*** (0.2401)	3.4460*** (0.0619)	3.0094(0.3302)
Access to credit	0.6602(0.7717)	0.1109(0.0100)	0.2389(0.0229)	0.0407(1.3183)	0.6402(0.0155)	0.3080(0.2007)	0.2109(0.0781)	0.4750(0.0019)
Farming Status	2.0652** (0.0013)	4.02345*** (0.0725)	3.1957*** (0.0053)	4.4421*** (0.0093)	0.0090(0.0161)	0.2100(0.3001)	0.22000(0.2700)	0.2177*** (0.6608)
Constant	5.04321*** (0.2113)	4.02347*** (0.0325)	3.4457*** (0.1653)	4.5121*** (0.1193)	5.2290*** (0.0861)	6.2191*** (0.3351)	3.8800*** (0.2713)	5.9077*** (0.3108)

Diagnosis based category

Number of observations	120
LR chi-square (50)	182.877***
Log Likelihood	-174.00290
R2	0.6260

Source: Field Survey, 2022

using zero or minimum tillage, use of diseases resistant, diversification and irrigation. Uloh, *et al*; (2023) opined that being fully engaged in farming, that the farmer will have enough time to explore more adaptation options and focus all the resources to farming as it the major source of income.

Table 4. Barriers to Adaptation to climate change

Variable	Frequency	Percentage
Poor access to credit	110	83.33
Lack of information	86	71.67
High cost of climate change adaptation	72	60.00
High cost of inputs	64	53.33
Inadequate labour	24	20.00

*Multiple Responses

Source: Field Survey; 2022

Table 4 showed the barriers to climate change adaptation as identified by the cassava farmers in the study area. The result revealed that 83.33% of the cassava farmers indicated that poor access to credit was the major barrier to climate change adaptation in the study area. This result is synonymous with the findings of Mabe; *et al*; (2014). They were of the view that high interest rate and modality of loan repayment may have deterred farmers from having access to credit especially in informal institution. Other variables identified as barrier to adaptation to climate include lack of information 71.67%, high cost of climate change adaptation 60.00% and high cost of labour input 53.33%. The least, was inadequate labour 20.00%. This may possibly be ascribed to the fact that, farmers get cheap labour from their household members (Al-Hassan, *et al*; 2013; Denkyirah; *et al*; 2017). This could entail that cassava farmers do not have to spend extra cost on labour to adapt to climate change.

Conclusion and Recommendations

Result of the study revealed that cassava farmers in the study area were relatively young. This implied that youths were involved in farming and cassava farming was not for only the elderly ones.

The finding of the multinomial logit model analysis further showed that the coefficients of farmers' farming status, farming experience and membership of organization were significant and positive to choice of adaptation strategies.

The result also showed that farmer's years of farming experience had significant and positive influence on choice of adaptation strategies options in climate change.

Since the result revealed that the farmers in the study areas were relatively young, it was recommended that government, non-governmental organization and policy maker should focus on policies geared towards social and infrastructural development of the rural area. This will prevent

rural urban migration of the young ones. It will also increase the involvement of the youth in farming with a consequent increase in food availability.

It was recommended that, policies that will encourage experienced farmer to remain in cassava farming, such as provision of improved input at subsidized rate should be intensified and carried out.

Relevant policies aimed at encouraging individual farmer to join farming organization/cooperative society to enable them pull their resources together and enjoy the benefits of economies of scale was also recommended.

Timely access by farmers to weather information through the office of metrological services was also recommended. This will in no little measure act as a guide to farmers in production decision making process in selecting adaptation options.

It was recommended that farmers should be encouraged to carry out production diversification. It is expected that this measure will help reduce or avert risks and uncertainties associated with agriculture as a result of climate change as well as a multi stream source of income to the farmers.

The need to enhance farmers' access to credit facility at reduced interest rate by government agency and Non-Governmental Organization concerned was recommended.

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